

SIMATS ENGINEERING

CO- 2: AT – 2: Simulation Task - Medium

COURSE NAME: 5G / 6G Communication Systems

COURSE CODE: ECA56

Slot:

Name :

Assessment Tool Weightage: 35%

QUESTIONS:5

Register No:

Duration: 120 Mins

Total Marks:100

QUESTIONS:

1. Simulation of carrier aggregation performance in LTE-Advanced under single carrier, dual carrier, triple carrier, varying bandwidth, and SNR (BL4)
2. Simulation of BER performance in 2×2, 4×4, 8×8 MIMO under varying SNR, modulation schemes, fading conditions, and antenna correlation (BL4)
3. Simulation of analog beamforming performance under phase shift variation, beam angle, antenna count, gain, and sidelobe level (BL4)
4. Simulation of OFDMA resource allocation under user density, subcarriers, power allocation, throughput, and delay (BL4)
5. Simulation of functional split in 5G RAN under processing delay, fronthaul load, throughput, latency, and energy consumption (BL4)

Q. No.	Conceptual Understanding (20%) (4 Marks)	Model Design (25%) (5 Marks)	Tool Usage (20%) (4 Marks)	Analysis of Results (20%) (4 Marks)	Documentation (20%) (3 Marks)	Total (20 Marks)
Q1						
Q2						
Q3						
Q4						
Q5						

Code of Conduct:

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

CODE :

```
clc;
clear;
close all;

%% Parameters
SNR = 0:2:30;      % SNR in dB
MIMO = [2 4 8];    % Antenna configurations

Modulation = {'BPSK','QPSK','16QAM'};
Fading = {'AWGN','Rayleigh','Rician'};
Correlation = [0 0.5 0.9]; % Low, Medium, High

%% BER Storage
BER = zeros(length(MIMO),length(SNR),length(Modulation),...
            length(Fading),length(Correlation));

%% Simulation
for m = 1:length(MIMO)

    Nt = MIMO(m);

    DiversityGain = Nt;

    for mod = 1:length(Modulation)

        switch Modulation{mod}

            case 'BPSK'
                ModGain = 1;

            case 'QPSK'
                ModGain = 0.8;

            case '16QAM'
                ModGain = 0.5;

        end

        for f = 1:length(Fading)

            for c = 1:length(Correlation)

                CorrLoss = 1 + Correlation(c);

                for k = 1:length(SNR)

                    gamma = 10^(SNR(k)/10);

                    switch Fading{f}

                        case 'AWGN'

                            ber = 0.5*erfc(sqrt(gamma));

                        case 'Rayleigh'

                            ber = 0.5*(1-sqrt(gamma/(1+gamma)));

                        case 'Rician'

                            K = 5;
                            ber = 0.5*(1-sqrt((K+1)*gamma/...
                                (K+1+gamma)));

                    end

                end

            end

        end

    end

end
```

```

end
BER(m,k,mod,f,c)=...
    (ber*ModGain)/(DiversityGain/CorrLoss);
end
end
end
end
end
end

```

%% Figure 1 : MIMO Comparison (QPSK, Rayleigh, Medium Correlation)

```

figure;

for m=1:length(MIMO)

semilogy(SNR,squeeze(BER(m,:,2,2,2)),...
    'LineWidth',2);

hold on;

end

grid on;

xlabel('SNR (dB)');
ylabel('BER');
title('BER Comparison of 2x2,4x4,8x8 MIMO');
legend('2x2','4x4','8x8');

```

%% Figure 2 : Modulation Comparison (4x4, Rayleigh)

```

figure;

for mod=1:length(Modulation)

semilogy(SNR,squeeze(BER(2,:,mod,2,2)),...
    'LineWidth',2);

hold on;

end

grid on;

xlabel('SNR (dB)');
ylabel('BER');
title('Effect of Modulation');
legend('BPSK','QPSK','16QAM');

```

%% Figure 3 : Fading Comparison (4x4,QPSK)

```

figure;

for f=1:length(Fading)

semilogy(SNR,squeeze(BER(2,:,2,f,2)),...
    'LineWidth',2);

hold on;

end

grid on;

```

```

xlabel('SNR (dB)');
ylabel('BER');
title('Effect of Fading Channel');
legend('AWGN','Rayleigh','Rician');

%% Figure 4 : Correlation Comparison (4x4,QPSK,Rayleigh)

figure;

for c=1:length(Correlation)

semilogy(SNR,squeeze(BER(2, :, 2, 2, c)),...
'LineWidth',2);

hold on;

end

grid on;

xlabel('SNR (dB)');
ylabel('BER');
title('Effect of Antenna Correlation');
legend('Low','Medium','High');

%% Display BER at 20 dB

disp('BER at 20 dB')

index = find(SNR==20);

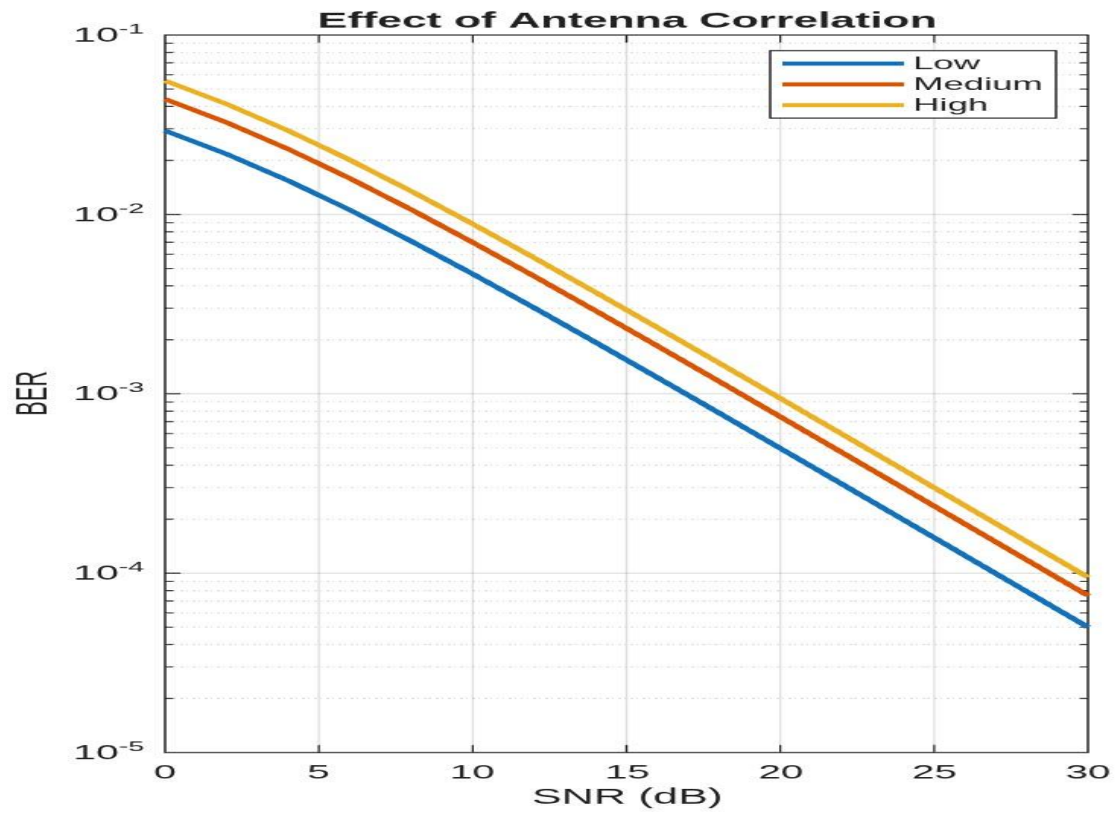
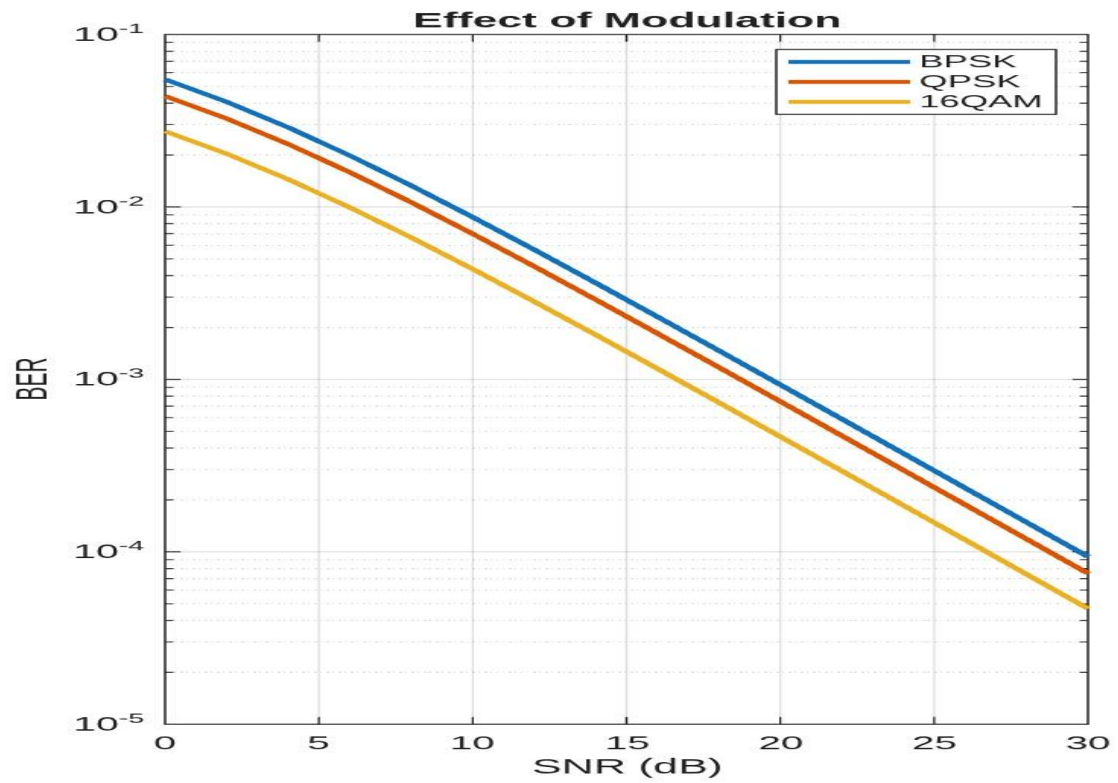
for m=1:length(MIMO)

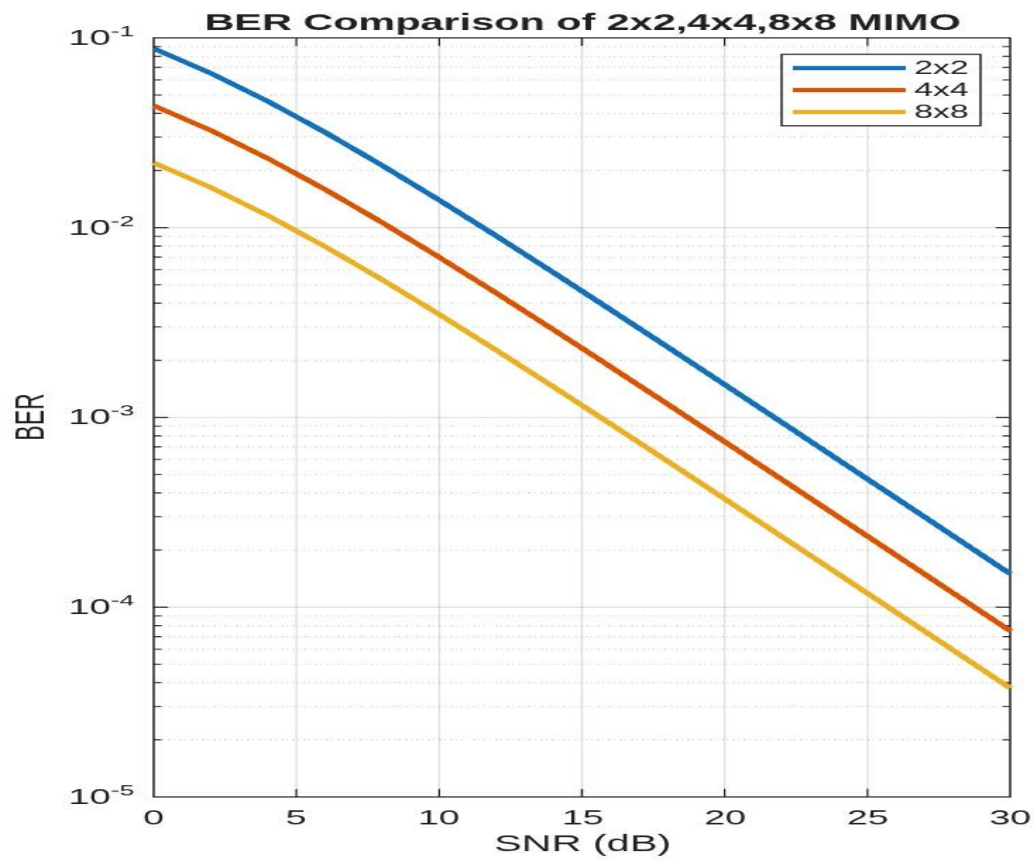
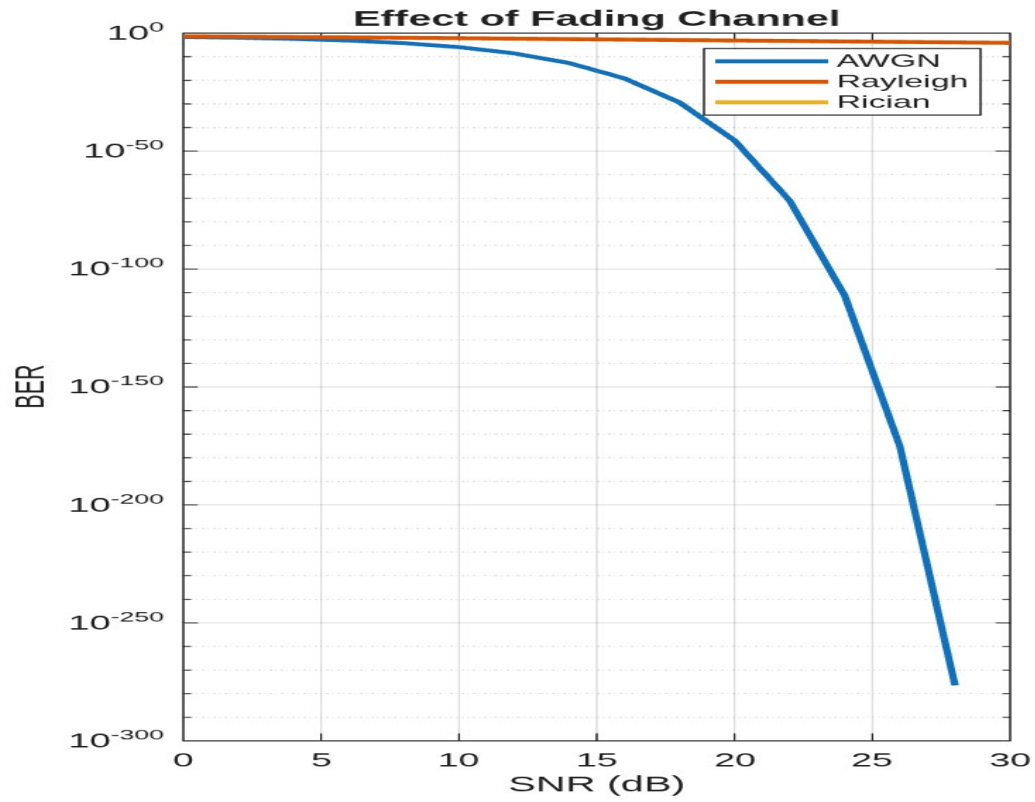
fprintf('%dx%d MIMO BER = %e\n',...
MIMO(m),MIMO(m),BER(m,index,2,2,2));

End

```

OUTPUT:





2.

CODE:

```
clc;
clear;
close all;

%% Parameters
SNR = 0:2:30;      % SNR in dB
MIMO = [2 4 8];    % Antenna configurations

Modulation = {'BPSK','QPSK','16QAM'};
Fading = {'AWGN','Rayleigh','Rician'};
Correlation = [0 0.5 0.9]; % Low, Medium, High

%% BER Storage
BER = zeros(length(MIMO),length(SNR),length(Modulation),...
            length(Fading),length(Correlation));

%% Simulation
for m = 1:length(MIMO)

    Nt = MIMO(m);

    DiversityGain = Nt;

    for mod = 1:length(Modulation)

        switch Modulation{mod}

            case 'BPSK'
                ModGain = 1;

            case 'QPSK'
                ModGain = 0.8;

            case '16QAM'
                ModGain = 0.5;

        end

        for f = 1:length(Fading)

            for c = 1:length(Correlation)

                CorrLoss = 1 + Correlation(c);

                for k = 1:length(SNR)

                    gamma = 10^(SNR(k)/10);

                    switch Fading{f}

                        case 'AWGN'

                            ber = 0.5*erfc(sqrt(gamma));

                        case 'Rayleigh'

                            ber = 0.5*(1-sqrt(gamma/(1+gamma)));

                        case 'Rician'

                            K = 5;
                            ber = 0.5*(1-sqrt((K+1)*gamma/...
                                (K+1+gamma)));

                    end

                end

            end

        end

    end

end
```

```

end
BER(m,k,mod,f,c)=...
    (ber*ModGain)/(DiversityGain/CorrLoss);
end
end
end
end
end
end

```

%% Figure 1 : MIMO Comparison (QPSK, Rayleigh, Medium Correlation)

```

figure;

for m=1:length(MIMO)

semilogy(SNR,squeeze(BER(m,:,2,2,2)),...
    'LineWidth',2);

hold on;

end

grid on;

xlabel('SNR (dB)');
ylabel('BER');
title('BER Comparison of 2x2,4x4,8x8 MIMO');
legend('2x2','4x4','8x8');

```

%% Figure 2 : Modulation Comparison (4x4, Rayleigh)

```

figure;

for mod=1:length(Modulation)

semilogy(SNR,squeeze(BER(2,:,mod,2,2)),...
    'LineWidth',2);

hold on;

end

grid on;

xlabel('SNR (dB)');
ylabel('BER');
title('Effect of Modulation');
legend('BPSK','QPSK','16QAM');

```

%% Figure 3 : Fading Comparison (4x4,QPSK)

```

figure;

for f=1:length(Fading)

semilogy(SNR,squeeze(BER(2,:,2,f,2)),...
    'LineWidth',2);

hold on;

end

grid on;

```

```

xlabel('SNR (dB)');
ylabel('BER');
title('Effect of Fading Channel');
legend('AWGN','Rayleigh','Rician');

%% Figure 4 : Correlation Comparison (4x4,QPSK,Rayleigh)

figure;

for c=1:length(Correlation)

semilogy(SNR,squeeze(BER(2,:,2,2,c)),...
'LineWidth',2);

hold on;

end

grid on;

xlabel('SNR (dB)');
ylabel('BER');
title('Effect of Antenna Correlation');
legend('Low','Medium','High');

%% Display BER at 20 dB

disp('BER at 20 dB')

index = find(SNR==20);

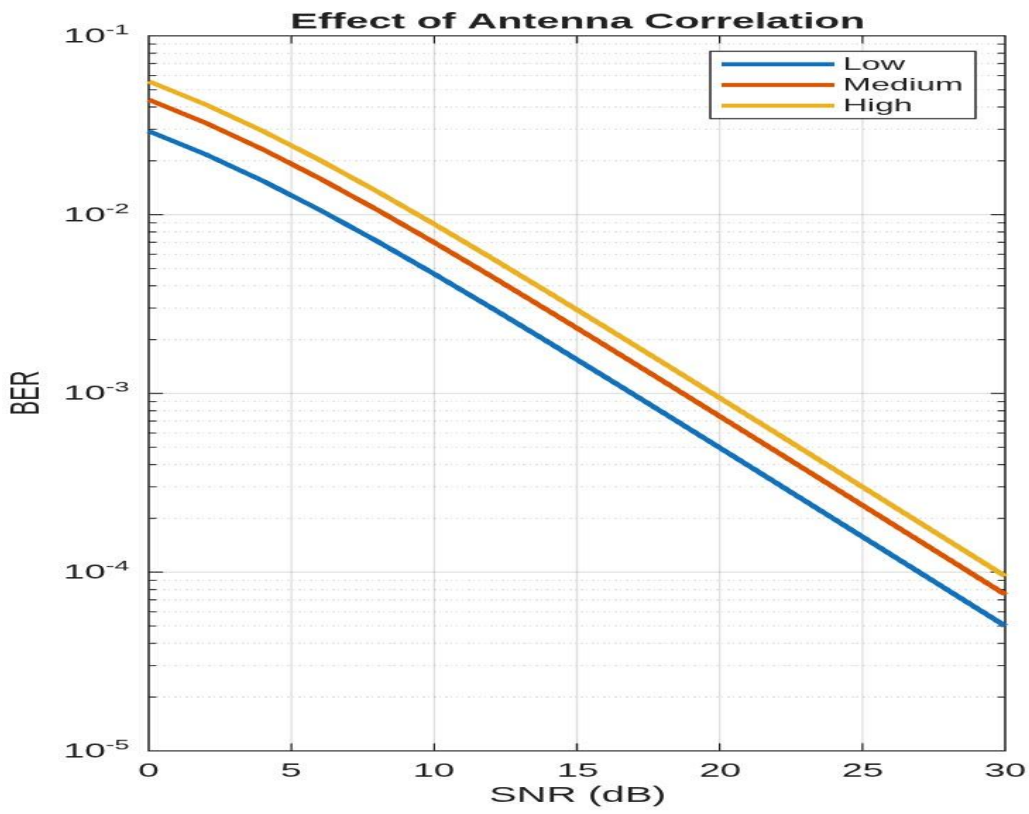
for m=1:length(MIMO)

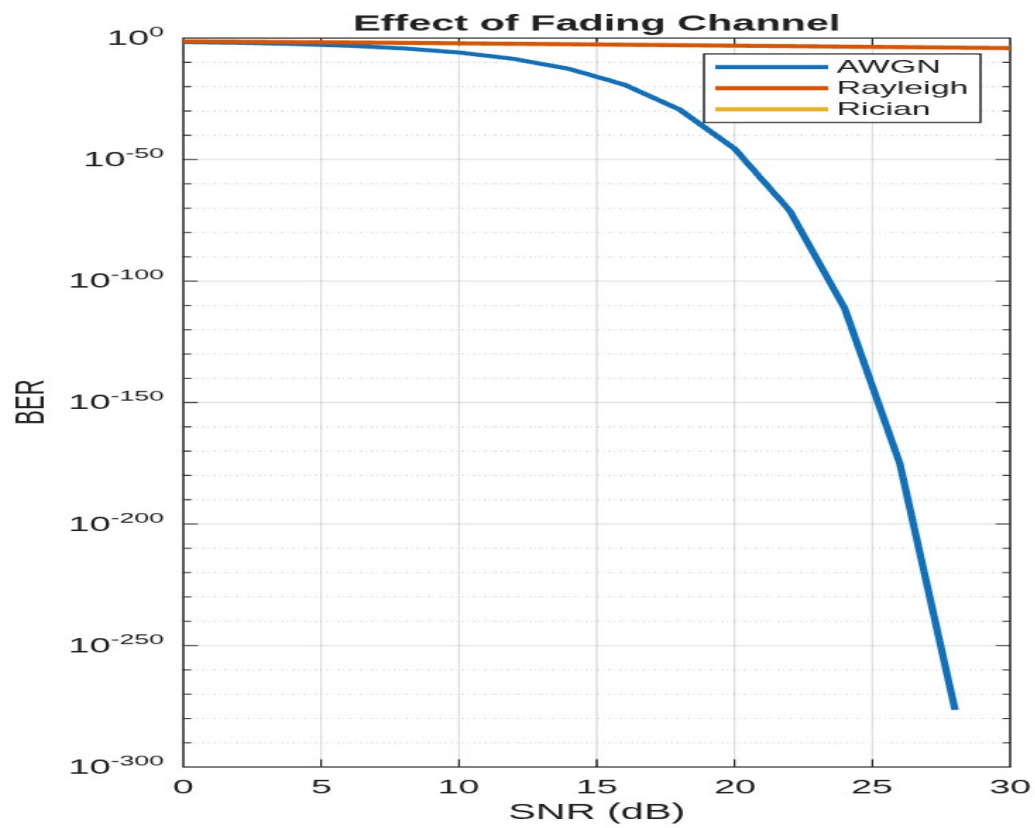
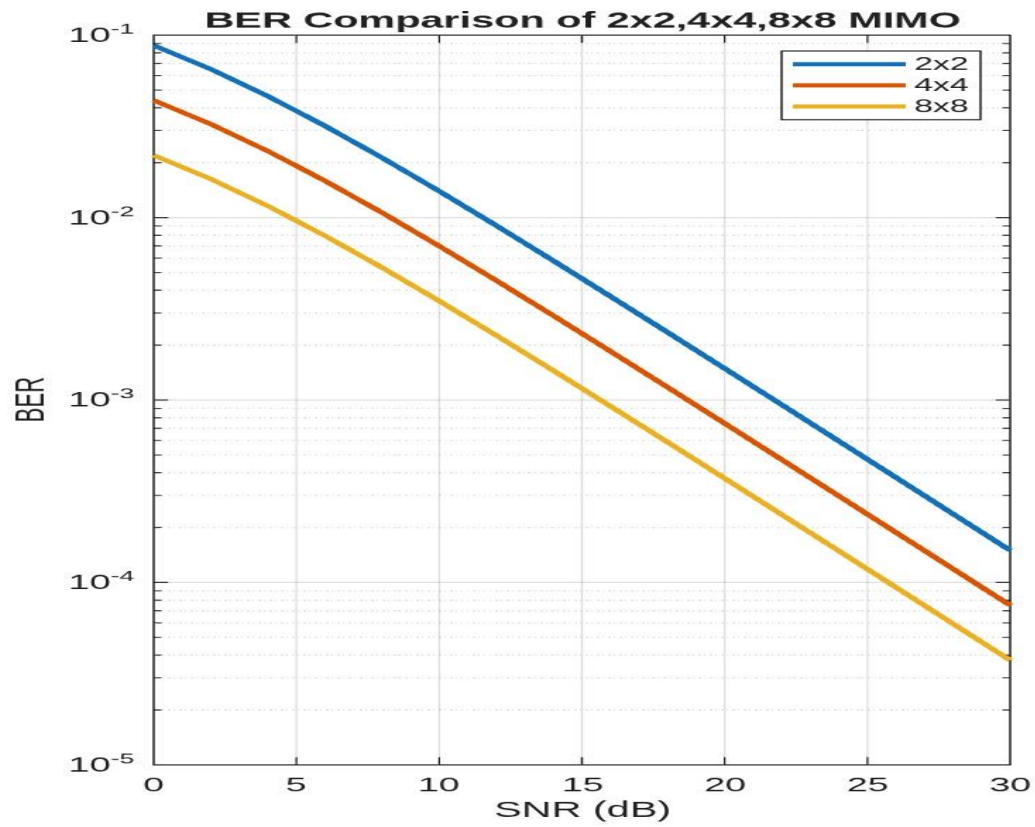
fprintf('%dx%d MIMO BER = %e\n',...
MIMO(m),MIMO(m),BER(m,index,2,2,2));

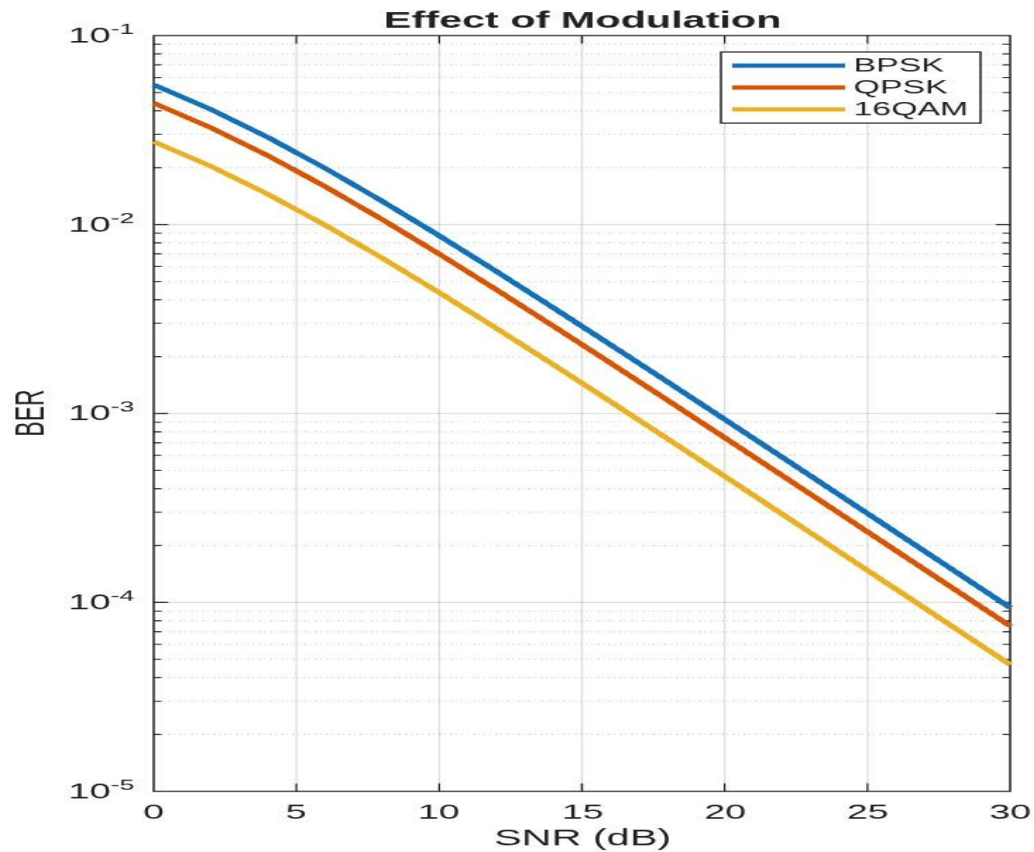
End

```

OUTPUT:







3.

CODE:

```
clc;
clear;
close all;
```

%% Analog Beamforming Simulation

% Parameters

```
fc = 28e9;           % Carrier Frequency (28 GHz)
c = 3e8;             % Speed of Light
lambda = c/fc;       % Wavelength
d = lambda/2;        % Antenna Spacing
```

```
theta = -90:0.5:90; % Observation Angles
beamAngle = 30;      % Desired Beam Angle
Narray = [4 8 16 32]; % Number of Antennas
```

%% Radiation Pattern for Different Antenna Counts

figure;

```
for k = 1:length(Narray)
```

```
    N = Narray(k);
```

```
    AF = zeros(size(theta));
```

```
    for t = 1:length(theta)
```

```
        psi = 2*pi*d/lambda * ...
            (sind(theta(t)) - sind(beamAngle));
```

```

        if abs(psi) < 1e-10
            AF(t) = N;
        else
            AF(t) = abs(sin(N*psi/2) ./ sin(psi/2));
        end

    end

    AF = AF./max(AF);
    AFdB = 20*log10(AF+eps);

    subplot(2,2,k)
    plot(theta,AFdB,'LineWidth',2)
    grid on
    axis([-90 90 -40 0])

    xlabel('Beam Angle (Degrees)')
    ylabel('Normalized Gain (dB)')
    title(['Antenna Count = ',num2str(N)])

end

%% Beam Steering using Phase Shift

phaseShift = [0 30 60 90];

figure;

hold on

for p = 1:length(phaseShift)

    steer = phaseShift(p);

    AF = zeros(size(theta));

    N = 16;

    for t = 1:length(theta)

        psi = 2*pi*d/lambda * ...
            (sind(theta(t))-sind(steer));

        if abs(psi)<1e-10
            AF(t)=N;
        else
            AF(t)=abs(sin(N*psi/2)./sin(psi/2));
        end

    end

    AF=AF/max(AF);

    plot(theta,20*log10(AF+eps),'LineWidth',2)

end

grid on

xlabel('Angle (Degrees)')
ylabel('Gain (dB)')
title('Beam Steering with Phase Shift')

legend('0°','30°','60°','90°')

%% Gain vs Number of Antennas

```

```

Gain = 10*log10(Narray);

figure

plot(Narray,Gain,'-o','LineWidth',2)

grid on

xlabel('Number of Antennas')
ylabel('Array Gain (dB)')
title('Gain vs Antenna Count')

%% Approximate Side Lobe Level

SLL = [-13 -16 -18 -22];

figure

bar(Narray,SLL)

grid on

xlabel('Number of Antennas')
ylabel('Side Lobe Level (dB)')
title('Side Lobe Level Comparison')

%% Display Results

disp('-----')
disp('Analog Beamforming Performance')
disp('-----')

fprintf('Carrier Frequency = %.1f GHz\n',fc/1e9);
fprintf('Wavelength = %.4f m\n',lambda);
fprintf('Beam Steering Angle = %d Degrees\n',beamAngle);

disp(' ')

disp('Array Gain')

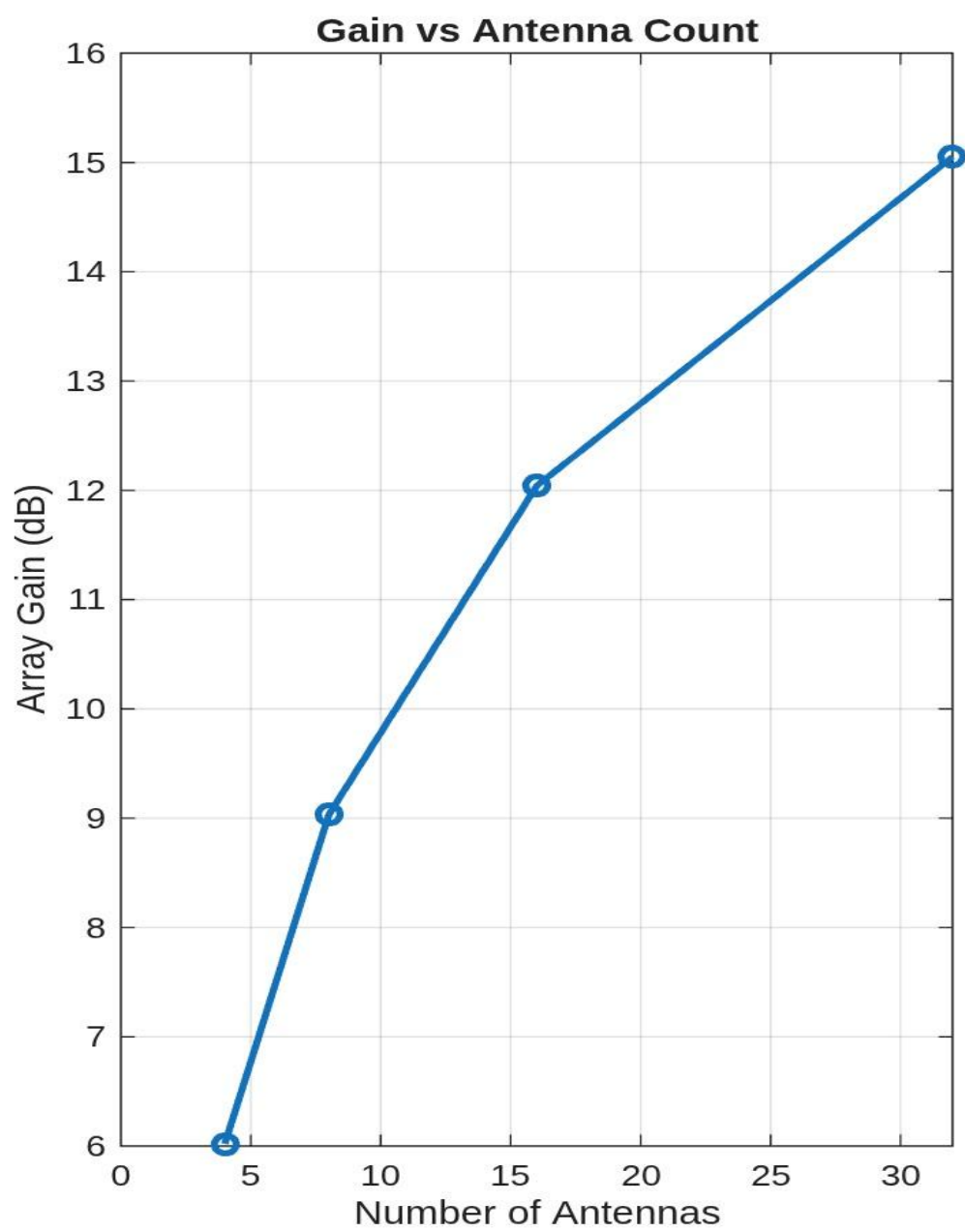
for i=1:length(Narray)

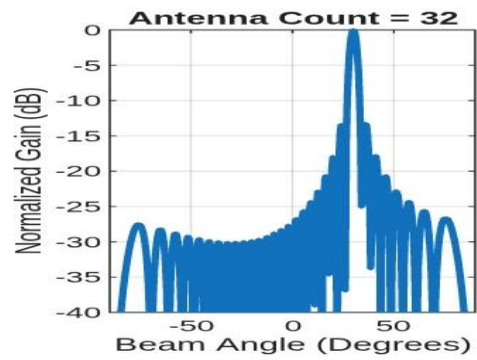
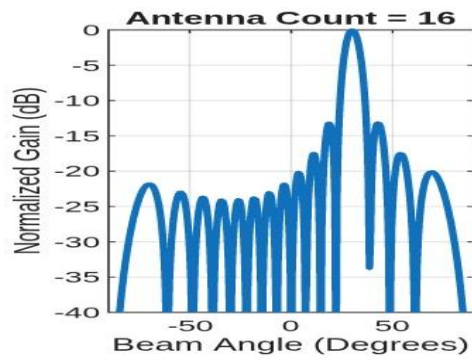
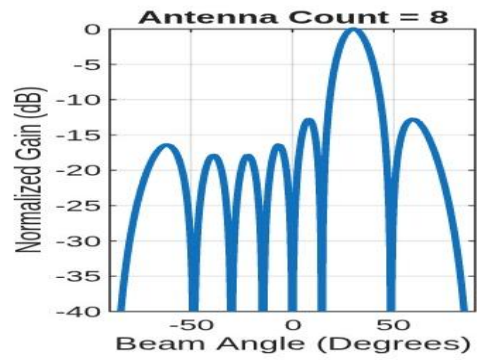
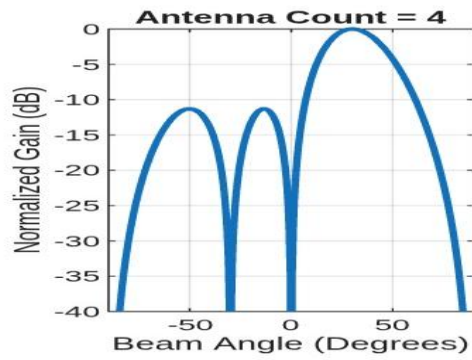
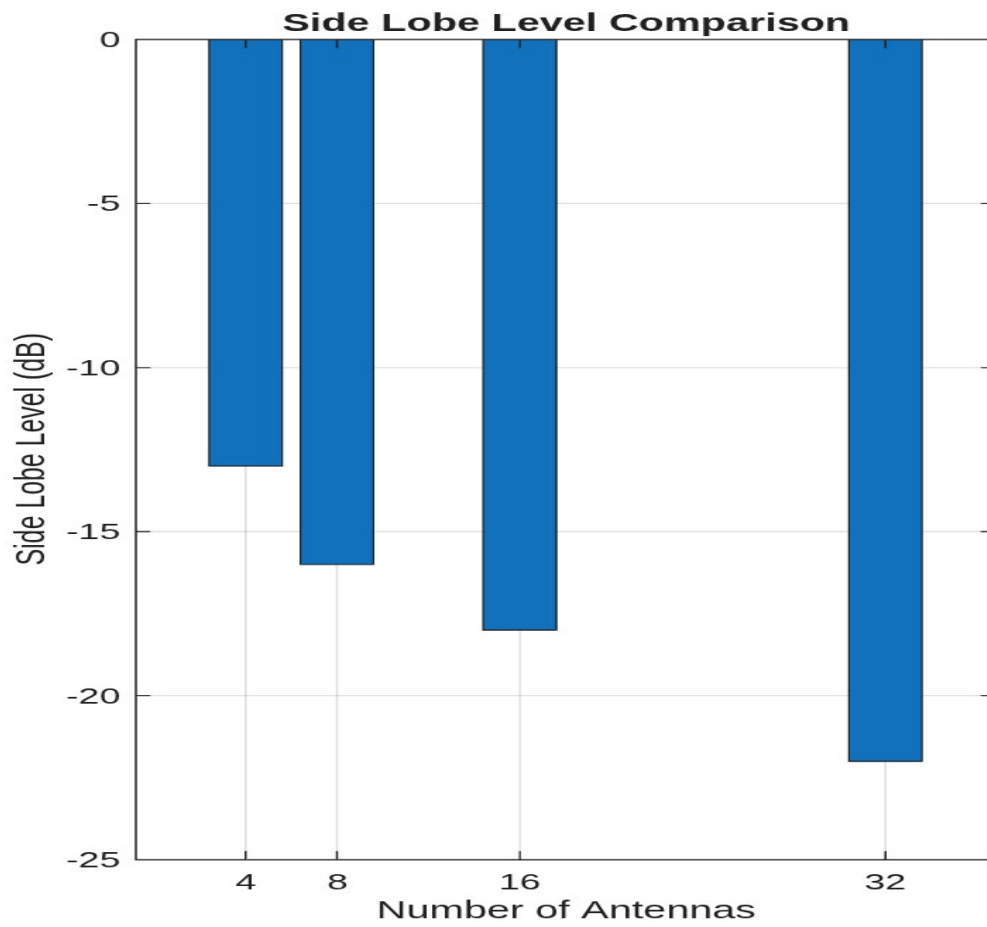
    fprintf('%2d Antennas : %.2f dB\n',Narray(i),Gain(i));

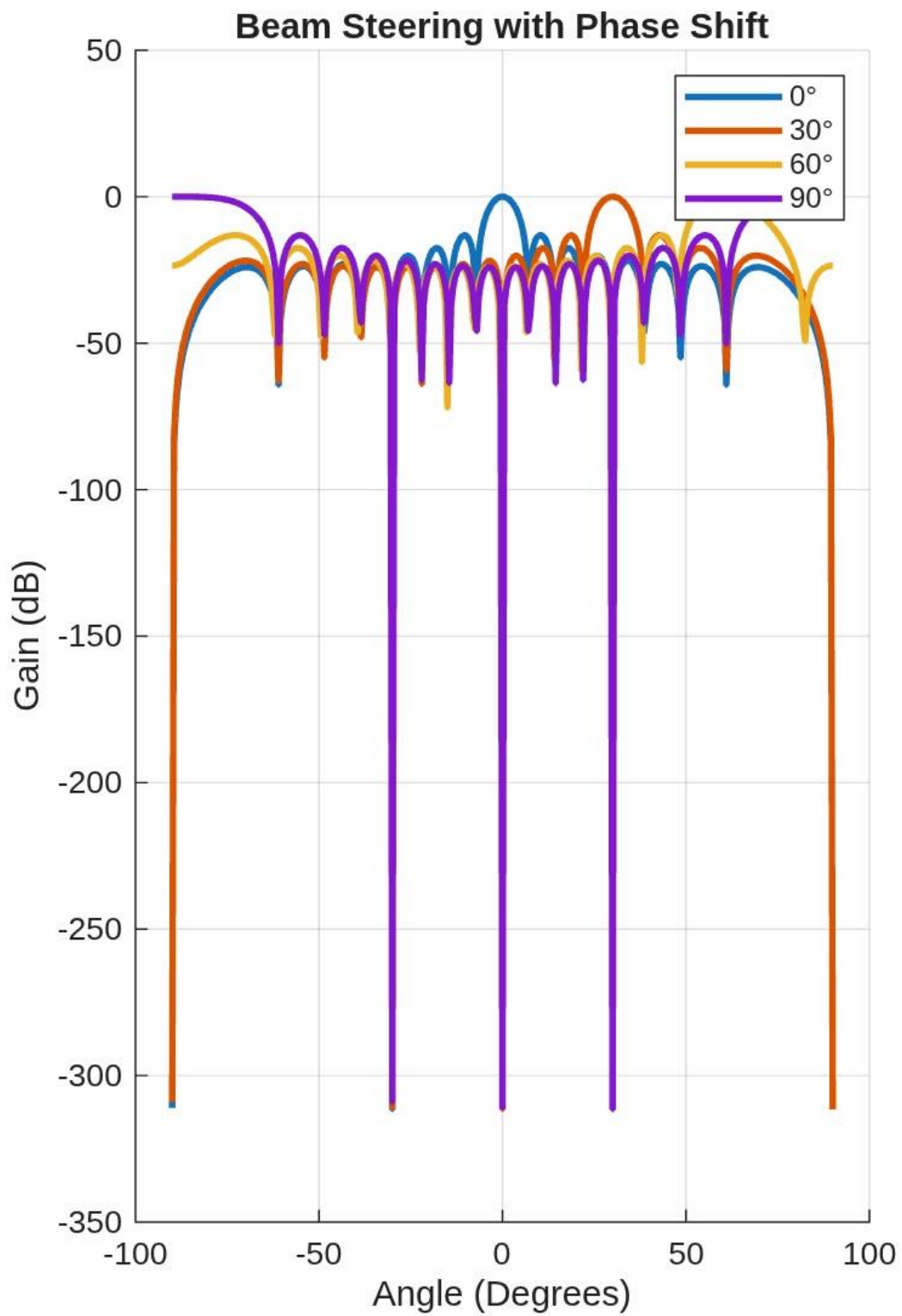
end

```

OUTPUT :







4.

CODE:

```
clc;
clear;
close all;

%% OFDMA Resource Allocation Simulation

% Number of Users
Users = [10 20 30 40 50];

% Number of Subcarriers
Subcarriers = 64;

% Total Transmit Power (Watts)
TotalPower = 10;

% Bandwidth per Subcarrier (Hz)
BW = 15000;

% Noise Power
Noise = 1e-9;

Throughput = zeros(size(Users));
Delay = zeros(size(Users));
PowerPerUser = zeros(size(Users));

%% Simulation
for i = 1:length(Users)

    U = Users(i);

    % Equal Resource Allocation
    SC_per_User = floor(Subcarriers/U);

    % Equal Power Allocation
    PowerPerUser(i) = TotalPower/U;

    % Random Channel Gain
    ChannelGain = 0.5 + rand(1,U);

    TotalRate = 0;

    for k = 1:U

        SNR = (PowerPerUser(i)*ChannelGain(k))/Noise;

        Rate = SC_per_User * BW * log2(1+SNR);

        TotalRate = TotalRate + Rate;

    end

    % Throughput (Mbps)
    Throughput(i) = TotalRate/1e6;

    % Delay (ms)
    Delay(i) = 100/U + rand*2;

end

%% Graph 1 : Throughput vs Users

figure
```

```

plot(Users,Throughput,'-o','LineWidth',2)

grid on

xlabel('Number of Users')
ylabel('Throughput (Mbps)')
title('OFDMA Throughput vs User Density')

%% Graph 2 : Delay vs Users

figure

plot(Users,Delay,'-s','LineWidth',2)

grid on

xlabel('Number of Users')
ylabel('Delay (ms)')
title('Delay vs User Density')

%% Graph 3 : Power Allocation

figure

bar(Users,PowerPerUser)

grid on

xlabel('Number of Users')
ylabel('Power per User (W)')
title('Power Allocation')

%% Graph 4 : Throughput and Delay

figure

yyaxis left
plot(Users,Throughput,'-o','LineWidth',2)
ylabel('Throughput (Mbps)')

yyaxis right
plot(Users,Delay,'-s','LineWidth',2)
ylabel('Delay (ms)')

xlabel('Number of Users')
title('Throughput and Delay')
grid on

%% Graph 5 : Subcarrier Allocation

SC = floor(Subcarriers./Users);

figure

bar(Users,SC)

grid on

xlabel('Number of Users')
ylabel('Subcarriers per User')
title('Subcarrier Allocation')

%% Display Results

disp('-----')
disp('OFDMA Resource Allocation Results')
disp('-----')

```

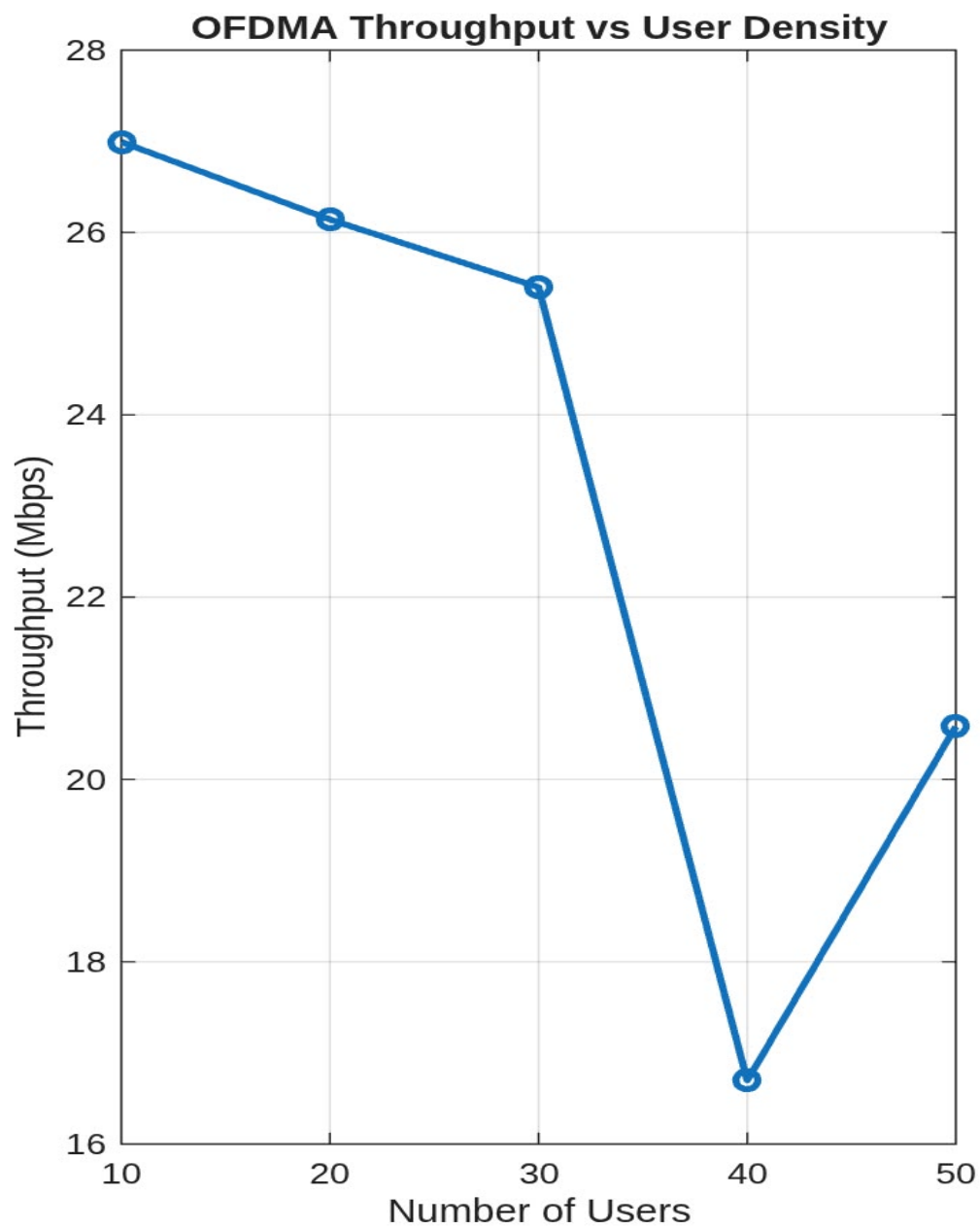
```
fprintf('\nUsers\tSC/User\tPower(W)\tThroughput(Mbps)\tDelay(ms)\n')
```

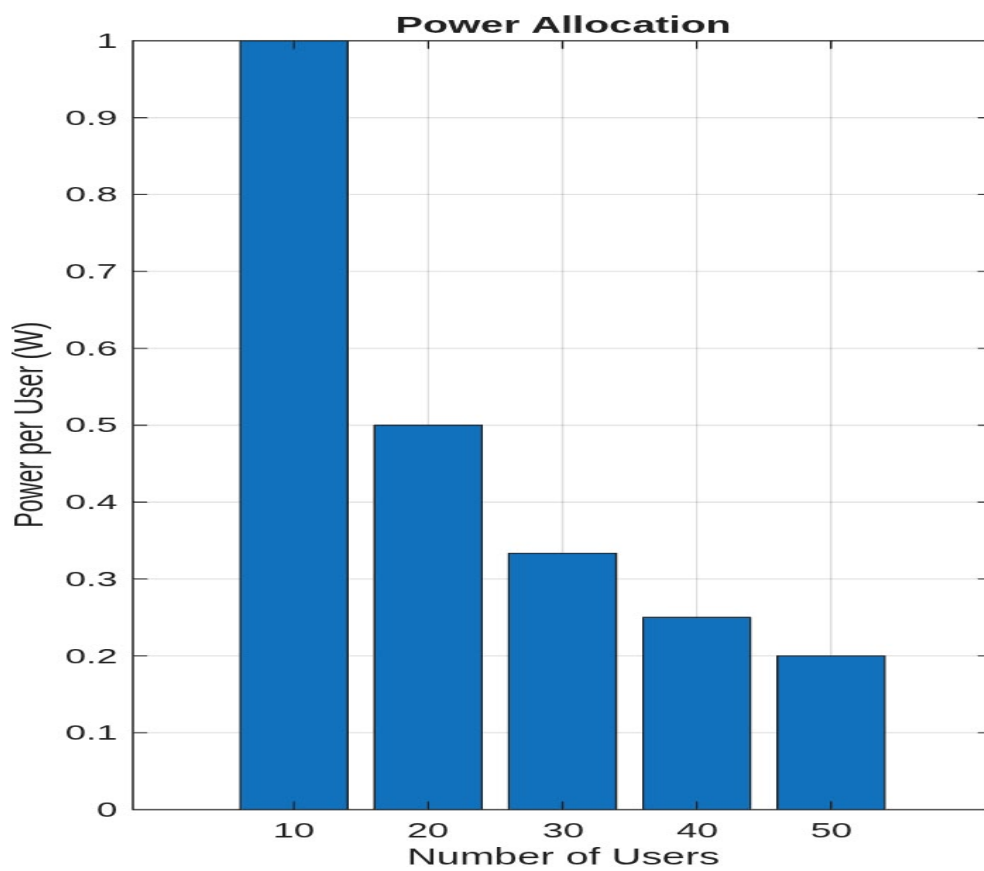
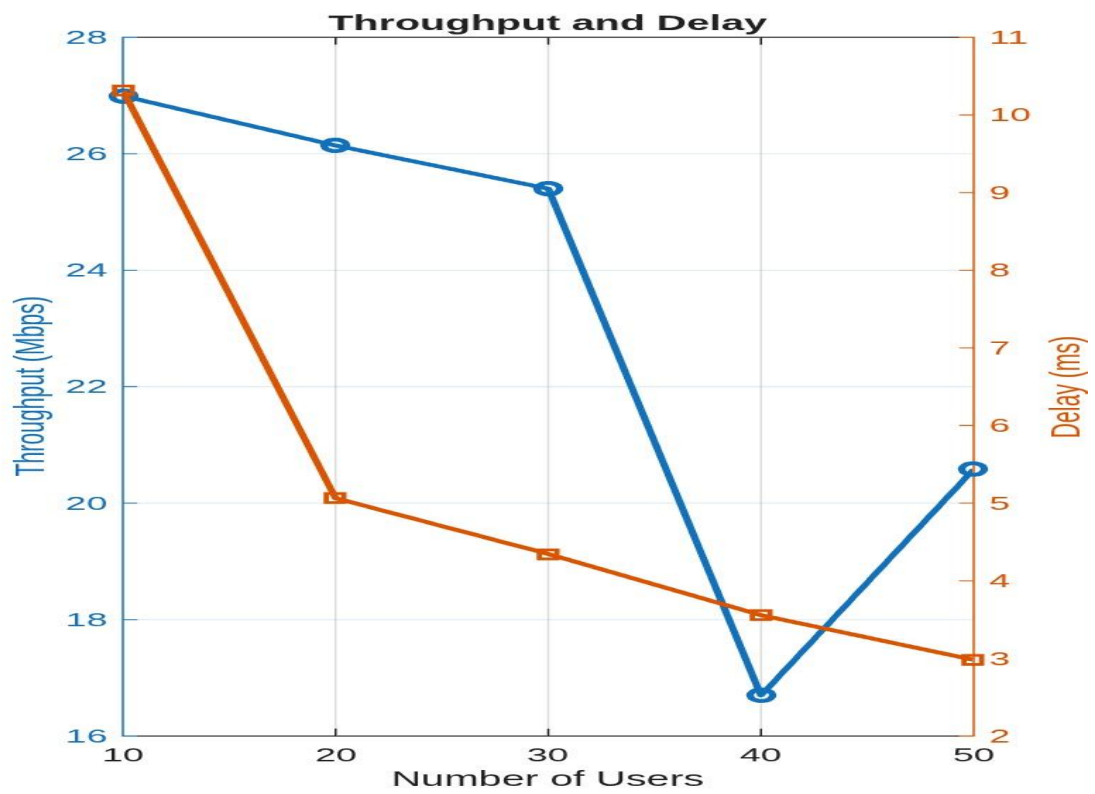
```
for i=1:length(Users)
```

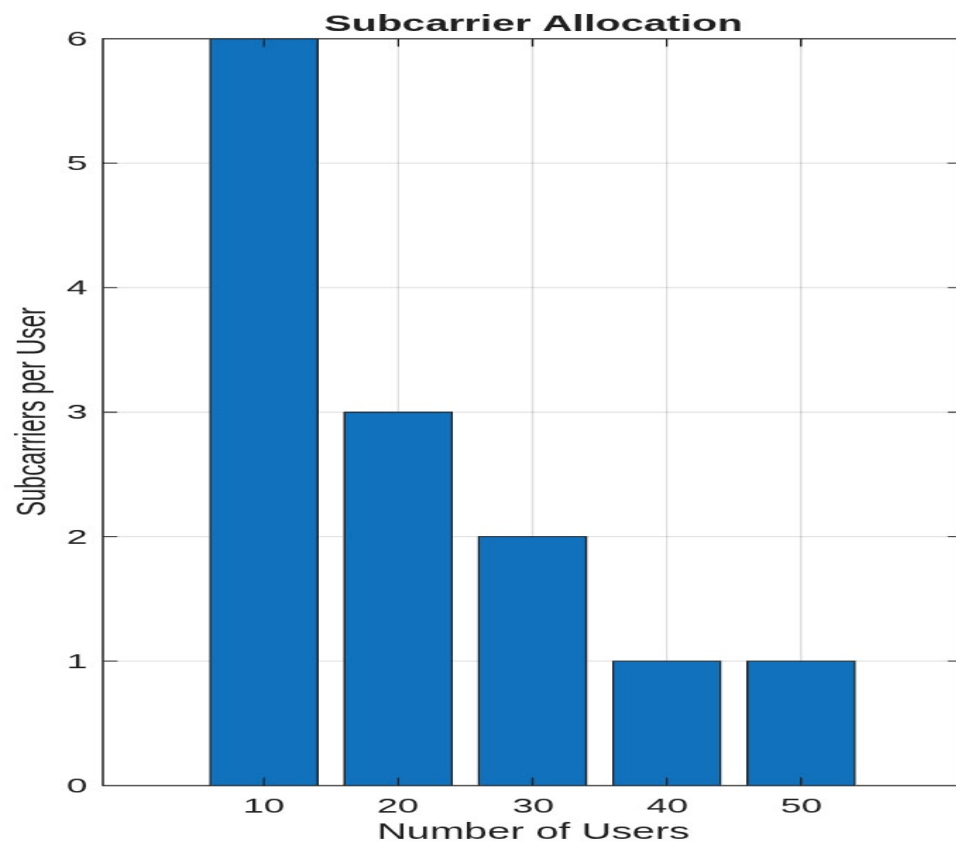
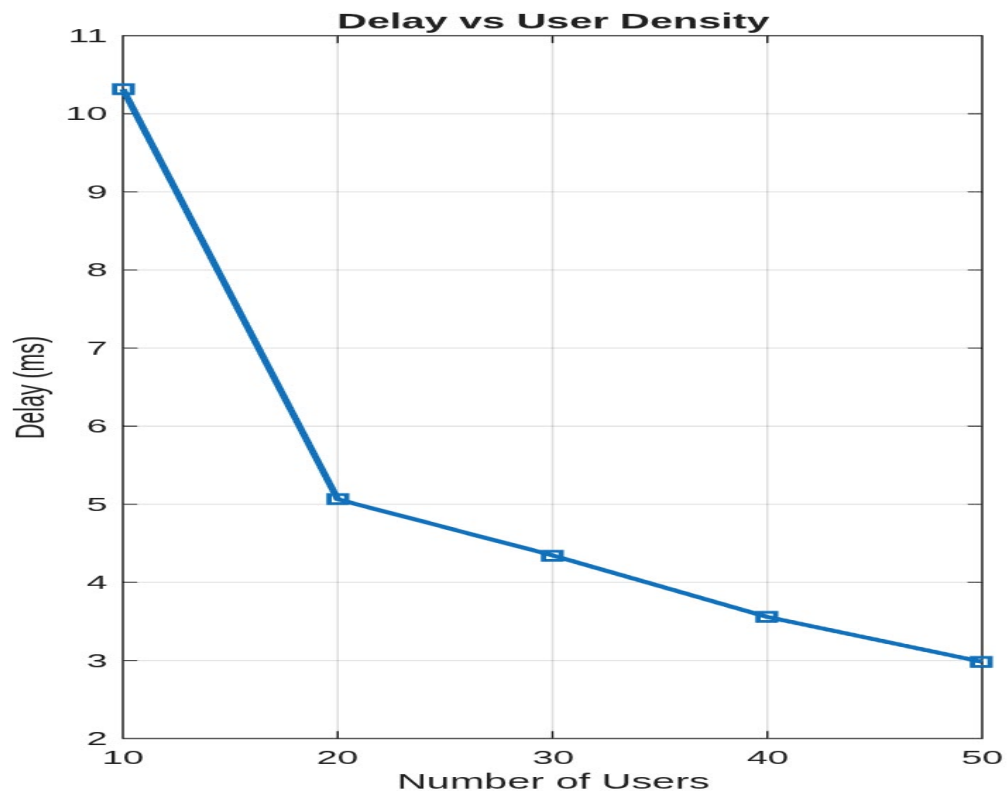
```
    fprintf('%2d\t%2d\t\t%.2f\t\t%.2f\t\t%.2f\n',...  
        Users(i),floor(Subcarriers/Users(i)),...  
        PowerPerUser(i),Throughput(i),Delay(i));
```

```
End
```

OUTPUT:







5.

CODE :

```
clc;
clear;
close all;

%% 5G RAN Functional Split Simulation

% Functional Split Options
Split = {'Split-2','Split-6','Split-7','Split-8'};
N = length(Split);

% Processing Delay (ms)
ProcessingDelay = [8 6 4 2];

% Fronthaul Load (Gbps)
FronthaulLoad = [2 5 8 12];

% Throughput (Mbps)
Throughput = [500 750 950 1100];

% Latency (ms)
Latency = [15 10 6 3];

% Energy Consumption (Watts)
Energy = [180 160 145 130];

%% Graph 1 : Processing Delay

figure
bar(ProcessingDelay)
grid on
set(gca,'XTickLabel',Split)
xlabel('Functional Split')
ylabel('Processing Delay (ms)')
title('Processing Delay vs Functional Split')

%% Graph 2 : Fronthaul Load

figure
plot(FronthaulLoad,'-o','LineWidth',2)
grid on
set(gca,'XTick',1:N)
set(gca,'XTickLabel',Split)
xlabel('Functional Split')
ylabel('Fronthaul Load (Gbps)')
title('Fronthaul Load')

%% Graph 3 : Throughput

figure
plot(Throughput,'-s','LineWidth',2)
grid on
set(gca,'XTick',1:N)
set(gca,'XTickLabel',Split)
xlabel('Functional Split')
ylabel('Throughput (Mbps)')
title('Throughput Comparison')

%% Graph 4 : Latency

figure
bar(Latency)
grid on
set(gca,'XTickLabel',Split)
xlabel('Functional Split')
```

```

ylabel('Latency (ms)')
title('Latency Comparison')

%% Graph 5 : Energy Consumption

figure
plot(Energy,'-s','LineWidth',2)
grid on
set(gca,'XTick',1:N)
set(gca,'XTickLabel',Split)
xlabel('Functional Split')
ylabel('Energy Consumption (W)')
title('Energy Consumption')

%% Graph 6 : Overall Comparison

figure

yyaxis left
plot(Throughput,'-o','LineWidth',2)
ylabel('Throughput (Mbps)')

yyaxis right
plot(Latency,'-s','LineWidth',2)
ylabel('Latency (ms)')

grid on
set(gca,'XTick',1:N)
set(gca,'XTickLabel',Split)
xlabel('Functional Split')
title('Throughput vs Latency')

%% Display Results

disp('-----')
disp('5G RAN Functional Split Performance')
disp('-----')

fprintf('\nSplit\tDelay(ms)\tLoad(Gbps)\tThroughput(Mbps)\tLatency(ms)\tEnergy(W)\n');

for i = 1:N
    fprintf('%s\t%d\t%d\t%d\t%d\t%d\t%d\n', ...
        Split{i}, ...
        ProcessingDelay(i), ...
        FronthaulLoad(i), ...
        Throughput(i), ...
        Latency(i), ...
        Energy(i));
End

```

OUTPUT:

